

ORIGINAL CONTRIBUTION

Early Recanalization After Tenecteplase or Alteplase in Distal and Medium Vessel Occlusion Strokes: An Observational Cohort Study

Nicolas Chausson¹ MD, PhD; Joseph Benzakoun¹ MD, PhD; Sofia Vargas-Ibarra² PhD; Yann L'Hermite¹ MD; Wagih Ben Hassen¹ MD, PhD; Didier Smadja¹ MD; Guillaume Turc¹ MD, PhD

BACKGROUND: Intravenous thrombolysis remains the standard treatment for distal and medium vessel occlusion (DMVO) strokes. However, unlike large vessel occlusions, whether tenecteplase achieves higher early successful recanalization rates than alteplase in DMVO strokes remains uncertain.

METHODS: This single-center observational cohort study included consecutive patients with magnetic resonance imaging (MRI)-confirmed DMVO stroke treated at Centre Hospitalier Sud-Francilien, France, according to an institutional thrombolysis protocol specifying alteplase (0.9 mg/kg, 2016–2018) or tenecteplase (0.25 mg/kg, 2018–2023), with follow-up MRI 1 to 2 hours after intravenous thrombolysis. Patients intended for endovascular thrombectomy or without diagnostic-quality follow-up MRI were excluded. Recanalization was blindly assessed using a modified Arterial Occlusion Lesion scale designed for MRI and DMVO. Early successful recanalization was defined as $\geq 50\%$ recanalization on early follow-up MRI (grade $\geq 2b$), and substantial early infarct growth, assessed using a semiautomatic segmentation technique, was defined as a ≥ 0.5 mL and $\geq 20\%$ increase in lesion volume. Propensity-score weighting accounted for baseline imbalances, including age, sex, vascular risk factors, baseline National Institutes of Health Stroke Scale score, occlusion site, and time period (2020–2023 versus 2016–2019).

RESULTS: Of the 319 included patients, 159 were treated with tenecteplase and 160 with alteplase. Median baseline National Institutes of Health Stroke Scale was 3 (1–7) in the tenecteplase group versus 4 (1–7.5) in the alteplase group, while M2 occlusions accounted for 54.1% versus 43.8% of cases. Early successful recanalization, assessed at a median postintravenous thrombolysis time of 80 (70–94) minutes, occurred in 72 (45.3%) and 53 (33.1%) patients treated with tenecteplase and alteplase, respectively (propensity-score weighting—odds ratio, 1.59 [95% CI, 1.15–2.20]; $P=0.005$). Substantial early infarct growth was observed in 42 (26.8%) and 64 (40.5%) patients, respectively (propensity-score weighting—odds ratio, 0.49 [95% CI, 0.35–0.69]; $P<0.0001$).

CONCLUSIONS: Compared with alteplase, tenecteplase-treated patients with DMVO had higher early successful recanalization and a lower risk of substantial early infarct growth, supporting its preferential use in this setting.

REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05635786.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: alteplase ■ cerebral infarction ■ stroke ■ thrombectomy ■ tenecteplase

Unlike large vessel occlusion (LVO) acute ischemic strokes (AIS), for which endovascular thrombectomy (EVT) is strongly beneficial,¹ 3 recent randomized trials do not support the use of EVT in patients with distal

and medium vessel occlusion (DMVO) AIS.^{2–4} Therefore, intravenous thrombolysis (IVT) remains the primary treatment strategy for DMVO-AIS in patients without contraindications. Although alteplase has long been the

Correspondence to: Nicolas Chausson, MD, PhD, Stroke Center, Sud-Francilien Hospital, 40 Ave Serge Dassault, 91100 Corbeil-Essonnes, France. Email nicolas.chausson@chsf.fr

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Nonstandard Abbreviations and Acronyms

AIS	acute ischemic stroke
AOL	Arterial Occlusive Lesion
DMVO	distal and medium vessel occlusion
DWI	diffusion-weighted imaging
ESR	early successful recanalization
EVT	endovascular thrombectomy
IVT	intravenous thrombolysis
LVO	large vessel occlusion
MRA	magnetic resonance angiography
MRI	magnetic resonance imaging
MRI-dAOL	MRI-distal Arterial Occlusive Lesion
MRI-H1	magnetic resonance imaging at 1–2 hours after treatment
NIHSS	National Institutes of Health Stroke Scale
PSW-OR	propensity-score-weighted odds ratio
SVS	susceptibility vessel sign
SWI	susceptibility-weighted imaging
TOF	time-of-flight

thrombolytic of choice for AIS, tenecteplase, a newer thrombolytic with greater fibrin specificity and a longer half-life, is now proven to be at least as safe and effective as alteplase.^{5–7} These results led to the recent approval of tenecteplase 0.25 mg/kg for AIS within 4.5 hours of symptom onset in Europe in January 2024 and thereafter in the United States.^{8,9} However, whether tenecteplase may actually be superior to alteplase remains debated.¹⁰

In the EXTEND-IA TNK trial, among patients with LVO-AIS scheduled for EVT, tenecteplase achieved significantly higher rates of early successful recanalization (ESR), a strong predictor of neurological recovery, compared with alteplase (22% versus 10%), which translated into better functional outcomes.¹¹ Whether such an advantage exists for patients with DMVO-AIS is still uncertain. This retrospective single-center observational cohort study aimed to compare ESR rates after IVT with tenecteplase or alteplase in patients with DMVO-AIS who underwent magnetic resonance imaging (MRI) both before and 1 to 2 hours after treatment (MRI-H1).

METHODS

Data Availability

The data that support the findings of this study are available from the corresponding author on reasonable request.

Study Design and Population

The DISTAL-IVT study (ClinicalTrials.gov identifier: REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05635786) was approved by the ethics

committee of the Centre Hospitalier Sud-Francilien (approval No. 01622). It is a single-center, retrospective, observational study conducted at Centre Hospitalier Sud-Francilien, a large primary stroke center in Corbeil-Essonnes, France, using data from a prospectively maintained stroke registry. This investigator-initiated study is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.¹² The completed [STROBE Checklist](#) is provided in the [Supplemental Material](#).

We reviewed all consecutive patients with AIS treated with IVT from March 2016 to December 2023 at Centre Hospitalier Sud-Francilien, where MRI is routinely used as the first-line imaging modality for suspected stroke, in accordance with French guidelines.¹³ As part of a potential sequential dual thrombolysis strategy (DIVARETRO study, REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05809921) for patients with DMVO-AIS, this center routinely performed follow-up MRI within 1 to 2 hours after IVT to assess early recanalization and, in selected cases, to consider an additional thrombolytic dose, as previously described.¹⁴

In the present study, we included all patients meeting the following eligibility criteria: age ≥ 18 years; MRI-confirmed AIS with DMVO as defined below; IVT within 4.5 hours of symptom recognition with either alteplase (0.9 mg/kg) or tenecteplase (0.25 mg/kg); and availability of a diagnostic-quality post-IVT MRI at 1 to 2 hours, including susceptibility-weighted imaging (SWI). Patients with unknown stroke onset were included in case of diffusion-weighted imaging (DWI)-fluid-attenuated inversion recovery mismatch. Patients intended for EVT based on the results of the baseline MRI were excluded. Baseline characteristics of patients who did not undergo early follow-up MRI are provided in [Table S1](#).

Study Treatment Groups

From March 2016 to January 2018, alteplase (0.9 mg/kg) was the standard thrombolytic drug at Centre Hospitalier Sud-Francilien. In early February 2018, following the presentation of the EXTEND-IA TNK trial results,¹¹ tenecteplase (0.25 mg/kg) replaced alteplase as the standard thrombolytic drug for all patients with AIS. However, alteplase was temporarily reintroduced for most patients during a tenecteplase shortage in Europe from March to September 2021 and again from the end of May 2022 until the end of the study period ([Figure S1](#)). Importantly, treatment allocation during these periods was driven solely by drug availability and not by clinical characteristics.

Clinical and Imaging Data

Clinical variables collected included age, sex, vascular risk factors, antithrombotic therapy, and National Institutes of Health Stroke Scale (NIHSS) scores at baseline and 1 hour after IVT.

Pre- and post-IVT MRI scans (baseline MRI and MRI-H1) were acquired on the same 1.5-T scanner (Signa HDxt, GE Healthcare) using a standardized protocol including DWI ($b=0$ and 1300 s/mm²; slice thickness, 3.6 mm), axial fluid-attenuated inversion recovery (TR/TE=11 000/127 ms; slice thickness, 4 mm), SWI (3-dimensional gradient-echo; slice thickness, 2 mm), and intracranial time-of-flight magnetic resonance angiography (TOF-MRA; slice thickness, 1.4 mm). All MRI assessments were independently reviewed by 2 trained

neurologists (D.S. and N.C.), blinded to the thrombolytic agent used, with discrepancies resolved by consensus.

DMVOs were defined as occlusions of distal M2-middle cerebral artery, nondominant proximal M2-middle cerebral artery, M3-middle cerebral artery, A2/A3-anterior cerebral artery, or P2/P3 posterior cerebral artery segments. Proximal dominant M2 or P1-posterior cerebral artery occlusions were also included if deemed ineligible for EVT, most commonly due to low NIHSS scores.

DMVOs were identified using TOF-MRA and the presence of a susceptibility vessel sign (SVS) on SWI. In patients with visible SVS, thrombus length was measured in 3 spatial planes on SWI, with the maximum value recorded.

Early recanalization was assessed using a modified Arterial Occlusion Lesion (AOL) scale that we adapted for DMVO evaluation on MRI-distal AOL (MRI-dAOL scale, Table 1). The MRI-dAOL scale was applied using TOF-MRA, SWI, and fluid-attenuated inversion recovery vascular hyperintensities in a hierarchical multimodal assessment strategy. Grading was based on direct comparison of baseline and follow-up MRI findings, taking into account interval changes in the site of occlusion as assessed across these modalities. In practice, TOF-MRA was primarily used to identify the presence and level of arterial occlusion, particularly for occlusions located in the more proximal part of the DMVO spectrum, and to guide subsequent clot assessment on SWI. When TOF-MRA was inconclusive, especially in more distal occlusions, SWI was used to identify the occlusive thrombus through the SVS and played a central role in MRI-dAOL grading by enabling assessment of clot presence, persistence, reduction, or distal migration on follow-up imaging. Fluid-attenuated inversion recovery vascular hyperintensities were used as complementary information to assess distal flow changes in the corresponding vascular territory. The final MRI-dAOL grade was assigned based on the overall consistency of the available multimodal MRI findings (see illustrative examples in Figure S2).

ESR was defined as an MRI-dAOL grade $\geq 2b$ on MRI-H1. DWI infarct growth was evaluated using voxel-based segmentation after coregistration of pre- and post-IVT DWI images, with volumetric analysis performed via a semiautomatic segmentation technique (Mango software, Research Imaging Institute, University of Texas; <http://rii.uthscsa.edu/mango/mango.html>).¹⁵ All measurements were performed by a single trained rater (N.C.) blinded to treatment group.

End Points

The primary end point was ESR, defined as an MRI-dAOL grade $\geq 2b$ on the follow-up MRI (MRI-H1; Table 1). Secondary efficacy radiological end points were complete early recanalization (MRI-dAOL grade 3), the degree of recanalization over the whole range of the MRI-dAOL scale on MRI-H1, change in thrombus length between baseline and MRI-H1, absolute change in DWI lesion volume between baseline and MRI-H1, and substantial early infarct growth. Substantial early infarct growth was defined as a $\geq 20\%$ relative increase and a ≥ 0.5 mL absolute increase in DWI volume between baseline and MRI-H1 (see Discussion). Clinical end points were change in NIHSS score between baseline and H1, and symptomatic intracranial hemorrhage at H1, defined as any intracranial hemorrhage associated with an NIHSS score increase of ≥ 4 points.

Statistical Analysis

Continuous variables were expressed as mean $\pm$ SD or median (interquartile range), and categorical variables as counts (%). Group comparisons used the χ^2 or Fisher exact test for categorical variables and the Mann-Whitney *U* test or Kruskal-Wallis test for continuous variables, as appropriate.

To address potential imbalances between the 2 treatment groups, we performed a propensity-score-weighted analysis using the entire study population. All baseline variables were considered for inclusion in the logistic model used to estimate the propensity score of each patient to be

Table 1. Rating Scale for Recanalization on MRI for Distal Arterial Occlusion: MRI-Distal Arterial Occlusive Lesion Scale Assessment

MRI-distal Arterial Occlusive Lesion scale	Definition	TOF-MRA	SVS on SWI	FVH
0	Primary occlusive lesion unchanged	No change	SVS unchanged or shorter but originating at the same location	No change
1	Thrombus debulking without recanalization (no distal flow)	Occluded arterial segment slightly longer, with no distal flow	Migration within the same arterial segment	No change
2a	Partial/complete recanalization of the primary occlusive lesion with residual occlusion in a major distal branch (recanalization $<50\%$ of the affected territory)	Incomplete occlusion or partial local recanalization with minor distal flow and appearance of a minor distal branch	Migration to a contiguous distal segment, or SVS decrease with appearance of a minor distal branch on TOF-MRA	FVH persisting in $>50\%$ of original territory
2b	Partial/complete recanalization of the primary occlusion with residual occlusion in a minor distal branch (recanalization $\geq 50\%$ of the affected territory)	Incomplete occlusion or partial local recanalization with major distal flow and appearance of a major distal branch	Migration to a noncontiguous distal segment, or to a contiguous distal segment with appearance of a major distal branch on TOF-MRA	FVH persisting in $\leq 50\%$ of original territory
2b+	Near-complete recanalization with residual distal thrombus	Aspect of complete recanalization	Distal thrombus migration	Minimal residual FVH
3	Complete recanalization of the primary occlusion with no residual thrombus and no residual FVH	Aspect of complete recanalization	No residual SVS	No residual FVH

FVH indicates fluid-attenuated inversion recovery vascular hyperintensities; MRI, magnetic resonance imaging; SVS, susceptibility vessel sign; SWI, susceptibility-weighted imaging; and TOF-MRA, time-of-flight magnetic resonance angiography.

treated with tenecteplase (as opposed to alteplase), considering potential multicollinearity. All baseline variables listed in Table 2 were included in the propensity-score model, except thrombus length, onset-to-IVT time, and anticoagulant use. Thrombus length was excluded because it could not be measured in patients without an SVS, onset-to-IVT time because it is intrinsically influenced by the shorter preparation and administration time of tenecteplase,¹⁶ and anticoagulant use because of its high correlation with a history of atrial fibrillation.

Balance of baseline characteristics between the 2 treatment groups was assessed before and after inverse probability of treatment weighting by calculating absolute mean standardized differences, with an absolute mean standardized difference $\leq 10\%$ indicating negligible imbalance. Associations between treatment (tenecteplase versus alteplase) and each binary or ordinal outcome were assessed using logistic regression models and expressed as propensity-score-weighted odds ratios (PSW-ORs) with 95% CIs. ESR rates stratified by occlusion site were explored. Sensitivity analyses were conducted with: (1) additional adjustment for onset-to-IVT time and (2) after exclusion of patients with proximal M2 occlusions. Analyses were conducted on available cases. No imputation was performed for missing data.

Interrater agreement for the primary end point was assessed with the weighted Cohen κ coefficient.¹⁷

All tests were 2-sided, with $P < 0.05$ considered statistically significant. Analyses were performed using SAS, version 9.4. G.T. and N.C. had full access to all the data in the study and assume responsibility for its integrity and for the data analysis.

RESULTS

The study flowchart is presented in the Figure. Among 574 consecutive patients with DMVO-AIS intended for IVT alone, 319 (55.6%) patients had an early follow-up MRI with interpretable images, of whom 159 (49.8%) were treated with tenecteplase and 160 (50.2%) with alteplase. The mean age was 72.2 ± 15.6 years, 163 (51.1%) patients were male, and the median NIHSS score on admission was 4 (1–7; Table 2). A total of 159 (49.8%) patients had an M2 occlusion. A visible thrombus on SWI (SVS) was present in 301 (94.4%) patients, with a median thrombus length of 7 (6–10) mm. The distributions of propensity scores according to treatment group are shown in Figure S3. Before PSW, only

Table 2. Baseline Characteristics and Process Times in Patients Treated With Tenecteplase or Alteplase

Variable	All (N=319)	Tenecteplase (n=159)	Alteplase (n=160)	ASMD before PSW	ASMD after PSW
Age, y	72.2 (15.6)	72.1 (14.6)	72.3 (16.5)	0	3
Male sex	163 (51.1%)	78 (49.1%)	85 (53.1%)	7	1
Period 2020–2023 (vs 2016–2019)	197 (61.8%)	103 (64.8%)	94 (58.8%)	11	0
Hypertension	184 (57.7%)	91 (57.2%)	93 (58.1%)	3	2
Diabetes	52 (16.3%)	28 (17.6%)	24 (15.0%)	4	1
Dyslipidemia	99 (31.0%)	43 (27.0%)	56 (35.0%)	19	1
Current smoking	44 (13.8%)	25 (15.7%)	19 (11.9%)	8	0
Previous stroke	47 (14.7%)	25 (15.7%)	22 (13.8%)	6	1
Coronary heart disease	40 (12.5%)	20 (12.6%)	20 (12.5%)	1	2
History of atrial fibrillation	35 (11.0%)	14 (8.8%)	21 (13.1%)	13	2
Antiplatelet therapy	96 (30.1%)	44 (27.7%)	52 (32.5%)	10	1
Anticoagulant therapy	30 (9.4%)	13 (8.2%)	17 (10.6%)	8	1
NIHSS score on admission	4.0 (1.0–7.0)	3.0 (1.0–7.0)	4.0 (1.0–7.5)	5	0
Intracranial occlusion site				16	0
Proximal M2-MCA	85 (26.6%)	46 (28.9%)	39 (24.4%)	...	...
Distal M2-MCA	71 (22.3%)	40 (25.2%)	31 (19.4%)	...	...
M3-M4-MCA	108 (33.9%)	54 (34.0%)	54 (33.8%)	...	...
ACA	11 (3.4%)	6 (3.8%)	5 (3.1%)	...	...
PCA	44 (13.8%)	13 (8.2%)	31 (19.4%)	...	...
Visible thrombus on SWI	301 (94.4%)	150 (94.3%)	151 (94.4%)	0	1
Thrombus length, mm	7.0 (6.0–10.0)	7.0 (6.0, 10.0)	8.0 (6.0–10.0)	9	6
DWI volume on admission, mL	1.1 (0.2–5.5)	1.0 (0.1–4.5)	1.3 (0.3–5.9)	9	5
Unknown onset time	33 (10.3%)	12 (7.6%)	21 (13.0%)	18	1
Onset-to-IVT time, min	167 (135.0–211.0)	160.0 (134.0–208.0)	180.0 (137.5–227.0)	18	14

Data are presented as N (%), mean (SD), or median (interquartile range). Missing data: DWI volume on admission, n=3 (<1%). ACA indicates anterior cerebral artery; ASMD, absolute standardized mean difference; DWI, diffusion-weighted imaging; IVT, intravenous thrombolysis; MCA, middle cerebral artery; MRI, magnetic resonance imaging; NIHSS, National Institutes of Health Stroke Scale; PCA, posterior cerebral artery; and SWI, susceptibility-weighted imaging.

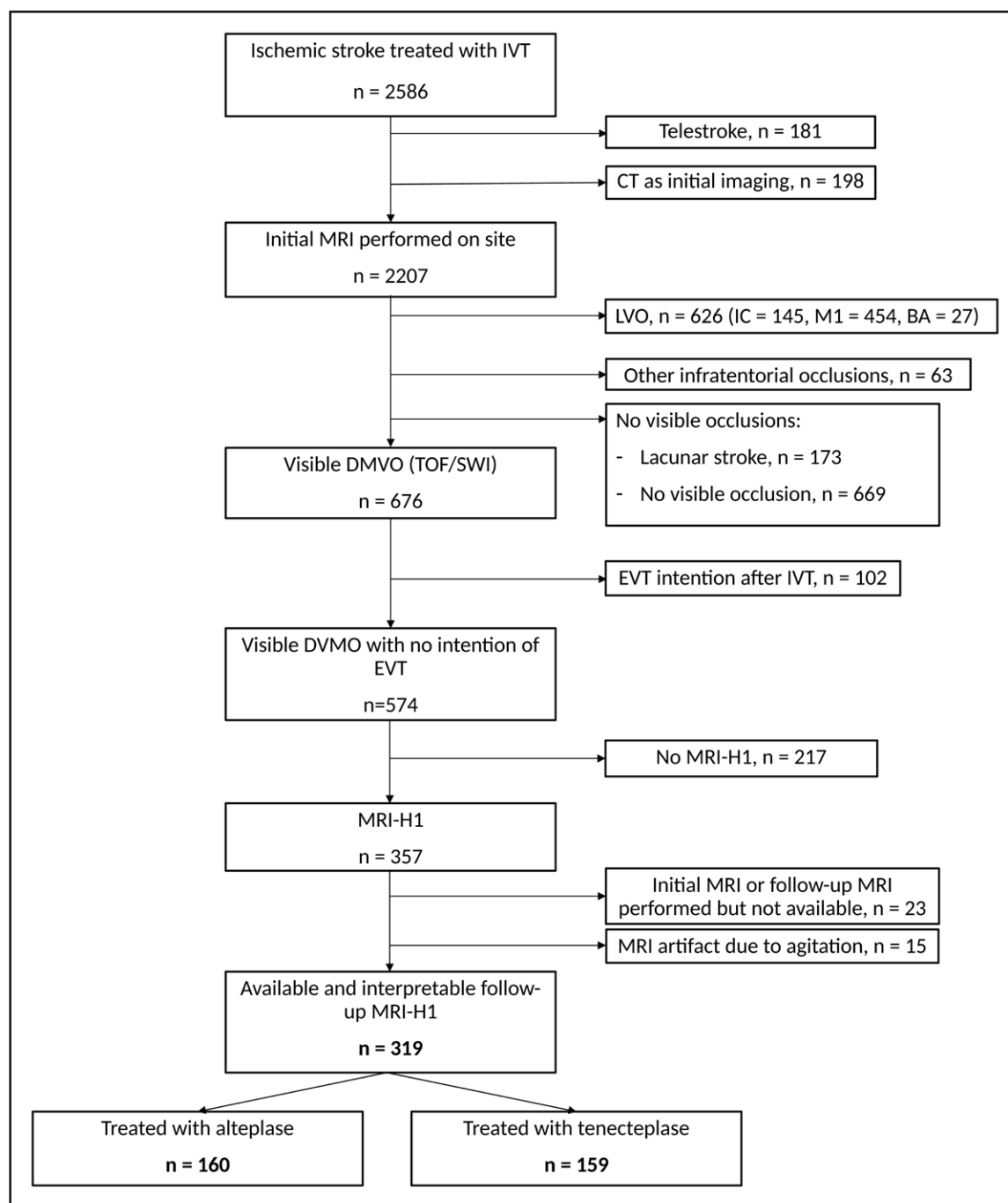


Figure. Flowchart

CT indicates computed tomography; DMVO, distal and medium vessel occlusions; EVT, endovascular thrombectomy; IVT, intravenous thrombolysis; LVO, large vessel occlusions; MRI, magnetic resonance imaging; SWI, susceptibility-weighted imaging; and TOF, time of flight.

moderate imbalances (absolute mean standardized differences, 0.11–0.20) in time period, dyslipidemia, atrial fibrillation, and occlusion site were observed. After PSW, the 2 treatment groups were well balanced, with absolute mean standardized differences now ≤ 0.10 (Table 2; Figure S4).

Onset-to-IVT time was shorter by 20 minutes in the tenecteplase group but the difference was not statistically significant (160.0 [134.0–208.0] minutes versus

180.0 [137.5–227.0] minutes; $P=0.12$). Median IVT-to-MRI-H1 time was 8 minutes shorter in the tenecteplase group (76.0 [66.0–90.0] minutes versus 84.5 [74.5–97.5] minutes; $P<0.0001$). Interobserver agreement for the MRI-dAOL scale was substantial, with weighted κ values of 0.75 (95% CI, 0.70–0.80) for scoring at 1 hour and 0.77 (95% CI, 0.71–0.85) for ESR assessment. Early efficacy and safety end points are presented in Table 3. ESR occurred in 72 (45.3%) and 53 (33.1%)

Table 3. Early Efficacy and Safety End Points

Variable	All (N=319)	Tenecteplase (n=159)	Alteplase (n=160)	Unadjusted effect estimates (95% CI)	Unadjusted P value	Adjusted effect estimates (95% CI)	Adjusted P value
ESR (MRI-dAOL ≥2b) at H1	125 (39.2%)	72 (45.3%)	53 (33.1%)	1.67 (1.06–2.63)	0.03	1.59 (1.15–2.20)	0.005
Complete early recanalization (MRI-dAOL 3) at H1	53 (16.6%)	34 (21.4%)	19 (11.9%)	2.02 (1.10–3.72)	0.02	1.92 (1.24–2.97)	0.003
MRI-dAOL on MRI-H1				1.60 (1.05–2.42)	0.04	1.66 (1.24–2.22)*	0.004
MRI-dAOL 0	165 (51.7%)	73 (45.9%)	92 (57.5%)	...	...	...	...
MRI-dAOL 1	13 (4.1%)	7 (4.4%)	6 (3.8%)	...	...	...	...
MRI-dAOL 2a	16 (5.0%)	7 (4.4%)	9 (5.6%)	...	...	...	...
MRI-dAOL 2b	33 (10.3%)	22 (13.8%)	11 (6.9%)	...	...	...	...
MRI-dAOL 2b+	39 (12.2%)	16 (10.1%)	23 (14.4%)	...	...	...	...
MRI-dAOL 3	53 (16.6%)	34 (21.4%)	19 (11.9%)	...	...	...	...
Decrease in thrombus length between baseline and H1, mm	1.0 (0.0 to 4.0)	2.0 (0.0 to 4.0)	1.0 (0.0 to 3.0)	NA	0.03	0.43 (−1.23 to 1.09)	0.20
Absolute change in DWI lesion volume between baseline and MRI-H1, mL	0.13 (−0.15 to 1.10)	0.03 (−0.22 to 0.80)	0.27 (−0.07 to 1.39)	NA	0.09	−0.30 (−0.54 to −0.06)†	0.01
Substantial early infarct growth (≥0.5 mL and ≥20%)	106/315 (33.7%)	42/157 (26.8%)	64/158 (40.5%)	0.54 (0.33 to 0.86)	0.01	0.49 (0.35 to 0.69)	<0.0001
Change in NIHSS score between baseline and H1	0.0 (0.0 to 2.0)	0.0 (0.0 to 2.0)	0.0 (0.0 to 2.0)	NA	0.94	0.08 (−0.55 to 0.72)	0.80
slCH at H1	2 (0.6%)	0 (0.0%)	2 (1.3%)	...‡	0.50‡	...§	...

Data are presented as N (%) or median (interquartile range). Adjusted effect estimates are shown as propensity-score-weighted odds ratios (PSW-ORs) with 95% CIs for binary and ordinal outcomes, and as propensity-score-weighted β coefficients (PSW- β) for continuous outcomes. Missing data: decrease in thrombus length between baseline and H1, n=6 (2%; among patients with a visible thrombus on baseline MRI); substantial early infarct growth, n=4 (1%). DWI indicates diffusion-weighted imaging; ESR, early successful recanalization; IVT, intravenous thrombolysis; MRI, magnetic resonance imaging; MRI-dAOL, magnetic resonance imaging-distal arterial occlusive lesion scale; NA, not available; NIHSS, National Institutes of Health Stroke Scale; and slCH, symptomatic intracranial hemorrhage.

*For MRI-dAOL categories, the adjusted estimate corresponds to a common odds ratio for better recanalization categories derived from ordinal logistic regression.

†Median regression.

‡Fisher exact test.

§No propensity-score weighting analysis performed due to no event in the tenecteplase group.

patients treated with tenecteplase and alteplase, respectively (PSW-OR, 1.59 [95% CI, 1.15–2.20]; $P=0.005$). Complete recanalization at 1 hour (MRI-dAOL, 3) was achieved in 34 (21.4%) patients receiving tenecteplase versus 19 (11.9%) with alteplase (PSW-OR, 1.92 [95% CI, 1.24–2.97]; $P=0.003$). Ordinal analysis of MRI-dAOL categories confirmed higher recanalization rates with tenecteplase (PSW-common OR for better recanalization categories, 1.66 [95% CI, 1.24–2.22]; $P=0.004$). Compared with alteplase, patients treated with tenecteplase had numerically higher recanalization rates at H1 for each initial occlusion site (Table S2). Reduction in thrombus size between baseline MRI and MRI-H1 was greater in patients treated with tenecteplase (2 [0–4] versus 1 [0–3] mm, $P=0.03$), but this difference did not remain significant after weighting (PSW- β , 0.43 [95% CI, −1.23 to 1.09]; $P=0.20$).

The absolute change in DWI lesion volume between baseline and MRI-H1 did not significantly differ between tenecteplase and alteplase in unadjusted analysis (0.03 mL [−0.22 to 0.80] versus 0.27 [−0.07 to 1.39]; $P=0.09$). However, this association reached statistical significance after PSW (PSW- β , −0.30 [−0.54 to −0.06]; $P=0.01$). Substantial early infarct growth was less frequently observed with tenecteplase (42 [26.8%] versus 64

[40.5%] patients; PSW-OR, 0.49 [95% CI, 0.35–0.69]; $P<0.0001$). Change in NIHSS score between baseline and H1 was 0 (interquartile range, 0–2) in both treatment groups (Table 3). Two patients treated with alteplase experienced symptomatic intracranial hemorrhage.

Sensitivity analyses reinforced the superiority of tenecteplase for ESR, with a stronger association after additional adjustment for symptom-to-thrombolysis delay (PSW-OR, 1.84 [95% CI, 1.30–2.60]; $P=0.0006$) or after excluding proximal M2 occlusions (N=286, PSW-OR, 2.14 [95% CI, 1.45–3.14]; $P=0.0001$).

DISCUSSION

This study provides new insights into early recanalization rates in patients with DMVO-AIS treated with IVT. We developed and applied a rating scale dedicated to the assessment of DMVO recanalization on MRI and showed substantial interobserver agreement. We found that tenecteplase was associated with significantly higher rates of ESR and lower rates of substantial early infarct growth compared with alteplase, as assessed by systematic MRI 1 to 2 hours after treatment. Higher recanalization rates with tenecteplase were observed across all occlusion sites, although these subgroup

differences were descriptive. This radiological advantage was achieved without notable safety concerns.

The lack of clinical benefit in recent randomized EVT trials for DMVO-AIS, along with a trend toward higher symptomatic intracranial hemorrhage rates, highlights the need to optimize medical reperfusion strategies, as IVT is currently the only effective option. While DMVO-AIS is generally associated with less severe clinical deficits than LVO-AIS, up to one-third to one-half of patients remain functionally disabled at 3 months.^{18,19} Optimizing treatment strategies and identifying the most effective thrombolytic agent for this population are therefore crucial. In the absence of proven benefit of EVT for DMVO-AIS, robust and reproducible noninvasive imaging scales are crucial.

Several scales have been proposed to assess cerebral recanalization after reperfusion therapy, the 2 most widely used being the Thrombolysis in Cerebral Infarction scale for angiography and the AOL scale for noninvasive imaging. However, these scales were developed in the context of LVO and rely on contrast-injection lumino-graphy. Although several modified versions of these scales have been proposed, their applicability to DMVO is limited. Indeed, Ospel et al²⁰ demonstrated that distal occlusions are challenging to identify on single-phase computed tomography angiography, the most widely used imaging modality in the acute phase, with an inter-rater agreement of only 58%. The recently proposed MeVO-modified Thrombolysis in Cerebral Infarction framework remains a purely DSA-based angiographic scale and therefore does not address the need for reliable noninvasive assessment in IVT-treated patients with DMVO.²¹ To date and to our knowledge, no adaptation of these scales has fully leveraged the multimodal information available from MRI. Applying the AOL scale to arterial blood flow imaging (ie, TOF-MRA) is challenging because it does not always allow the identification of the precise site of DMVO occlusion. However, the visualization of the DWI lesion and of potential fluid-attenuated inversion recovery vascular hyperintensities, along with the frequent detection of the thrombus on high-resolution SWI, allows a more focused analysis of the TOF-MRA and facilitates identification of the occlusion site. The MRI-dAOL scale developed in our study showed substantial interrater agreement for ESR, but further validation in independent DMVO cohorts imaged on different MRI platforms is warranted.

Our main results expand on prior findings from the EXTEND-IA TNK trial, which demonstrated higher ESR rates with tenecteplase compared with alteplase in LVO-AIS before EVT, to the context of DMVO-AIS, with an identical absolute difference of 12% across both settings.²² Data on early recanalization after IVT in DMVO-AIS are scarce. Ospel et al¹⁸ reported a 47.2% recanalization rate (75/159) with alteplase, assessed by contrast-enhanced computed tomography imaging at a

median of 6 hours post-IVT. In our cohort, tenecteplase achieved comparable rates within a median IVT-to-follow-up-imaging time of only 80 minutes. Ospel et al further observed that recanalization rates with alteplase continued to rise gradually, reaching ≈65% at 24 hours. This underscores the importance of considering IVT-to-follow-up imaging time when interpreting recanalization rates.

To our knowledge, only 1 secondary analysis of the AcT trial, including 455 patients, has compared tenecteplase and alteplase in DMVO-AIS.¹⁹ Within this cohort, a subgroup analysis of 87 patients who underwent EVT after IVT provided comparative data on early recanalization between the 2 treatments. Early pre-EVT recanalization was nearly twice as frequent with tenecteplase as with alteplase (13.0% versus 7.5%), although this difference did not reach statistical significance, likely due to limited statistical power. These findings are consistent with our results and further support the radiological advantage of tenecteplase in terms of early reperfusion.

Whether this radiological benefit translates into improved functional outcomes remains uncertain. In the same AcT analysis, which included all 455 patients with DMVO-AIS treated with either tenecteplase or alteplase, no significant difference was observed in 90-day outcomes (modified Rankin Scale, 0–1: 37.9% versus 34.7%; adjusted risk ratio, 1.07 [95% CI, 0.91–1.25]). The absence of a clinical difference likely reflects limited statistical power and the reduced sensitivity of the functional end point assessed by the modified Rankin Scale in patients with relatively mild baseline deficits.

By assessing changes in DWI lesion volume at 1 hour, a complementary surrogate marker of favorable outcome alongside ESR, our study provides additional radiological insights. We did not observe a significant difference in the absolute change in DWI lesion volume between baseline and MRI-H1 across treatment groups. This metric likely masks distinct lesion trajectories, expansion, partial reversal, or stability, which limits its interpretability. To better capture meaningful early infarct growth, we therefore adopted an a priori dual-threshold definition combining absolute and relative volume change. This approach minimizes the risk of interpreting segmentation variability as true infarct progression, particularly in patients with small ischemic lesions or minor changes, where the likelihood of individual misclassification is higher. For lesions <5 mL, previous studies have shown that a minimal change of 0.3 mL is required to mitigate segmentation-related errors, whereas for larger lesions, this absolute threshold is insufficient and must be complemented by a relative change criterion.^{23,24} Using our predefined dual-threshold definition (≥0.5 mL and ≥20% infarct growth), we observed significantly lower rates of substantial early infarct growth in the tenecteplase group compared with alteplase (26.8% versus 40.5%; $P<0.0001$). We hypothesize that tenecteplase achieved

faster and more effective recanalization, thereby shortening ischemia duration and limiting infarct expansion, consistent with findings previously reported in LVOs.²⁵

Altogether, our findings support tenecteplase over alteplase in DMVO-AIS, consistent with evidence already established in LVO before EVT, and reinforce the rationale for its broader adoption as the thrombolytic of choice in AIS. Nevertheless, the rate of early recanalization remained below 50%, underscoring the need for further therapeutic optimization in DMVO. In our study, among patients without early recanalization, approximately two-thirds showed no substantial infarct growth at 1 hour, suggesting a potential therapeutic window for additional intervention. Given the suboptimal efficacy of current IVT and the neutral results of recent distal thrombectomy trials, alternative approaches such as sequential dual intravenous thrombolysis warrant further investigation. We are currently preparing a multicenter, randomized, double-blind clinical trial, funded by the French Ministry of Health, to evaluate this approach. The primary end point will be functional outcome, while the rate of recanalization at 24 hours, assessed by MRI-dAOL, will serve as a key secondary end point. Other research teams have also initiated similar dual intravenous thrombolysis strategies for DMVOs, further supporting the growing interest in this therapeutic concept.²⁶

Our study has several strengths. First, it is based on unique radiological data, obtained from routine early MRI performed 1 hour after IVT in patients with DMVO-AIS, as part of the nonrandomized DIVARETRO study.¹⁴ Second, radiological evaluation was conducted with methodological rigor, in a blinded manner and with high interrater agreement. These radiological outcomes are recognized surrogate markers, closely correlated with clinical prognosis, and require fewer patients than trials based on long-term functional outcomes.²⁷

However, several limitations should be acknowledged. First, this was a retrospective, single-center observational study, which may introduce bias. However, clinical data were obtained from prospectively maintained databases, reducing the risk of information bias. Second, only 60% of patients with DMVO-AIS underwent 1-hour post-IVT MRI, as follow-up imaging was performed selectively based on the treating physician's judgment. The reasons for not obtaining follow-up MRI were not systematically documented but may have included subtle fluid-attenuated inversion recovery hyperintensity on baseline MRI or borderline indications for IVT, such as minor or nondisabling deficits, which could have excluded these patients from consideration for adjunctive therapies and, consequently, from early posttreatment MRI monitoring. Nevertheless, most baseline characteristics were well balanced between the 2 treatment groups even before propensity-score weighting, supporting clinical comparability and

consistent candidate selection over time, thereby minimizing confounding and ensuring a valid comparison. Third, DWI lesion volumes were assessed using a semi-automated segmentation approach based on visual inspection rather than a strict ADC-defined threshold, as early lesions were often subtle and poorly defined on ADC maps. Relying on a strict ADC-defined threshold could have led to underestimation of infarct size, particularly in small-volume strokes, which predominate in DMVO. However, as the assessor was blinded to treatment allocation, differential misclassification is unlikely. Finally, because some patients received additional treatment (ie, second-dose IVT) after the 1-hour MRI, 24-hour or 3-month radiological and clinical outcome comparisons between the tenecteplase and alteplase groups could not be interpreted.

CONCLUSIONS

Given the neutral results of recent randomized EVT trials in DMVO-AIS, optimizing IVT has become a critical therapeutic priority. Our results demonstrate that tenecteplase is associated with higher ESR and a lower incidence of substantial early infarct growth in DMVO-AIS compared with alteplase, adding to the growing body of evidence supporting tenecteplase as the preferred thrombolytic agent in this setting. However, these findings should be interpreted in light of the retrospective design, potential secular time confounding, and exclusion of EVT-intended patients. Future prospective studies are warranted to confirm these findings and to further refine therapeutic strategies for DMVO-AIS.

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Affiliations

Stroke Unit, Hôpital Sud-Francilien, Corbeil-Essonnes, France (N.C., Y.L.H., D.S.). Institute of Psychiatry and Neuroscience of Paris (IPNP), INSERM U1266, France (N.C., J.B., W.B.H., D.S., G.T.). Neuroradiology, GHU Paris Psychiatrie et Neurosciences; Université Paris Cité, France (J.B., W.B.H.). IBISC, Université Evry, Université Paris-Saclay, France (S.V.-I.). Neurology Department, GHU Paris Psychiatrie et Neurosciences; Université Paris Cité, France (G.T.). Université Paris-Saclay, France (N.C., Y.L.H., D.S.).

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Supplemental Material

Tables S1–S2
Figures S1–S4
STROBE Checklist

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